

Chapter 1 Homework Answer Sheet

Part A Multiple choice (10 points)

1-5 CDEBA

6-10 DBCAE

Part B Short questions (10 points)

11. (2 points)

$$\begin{aligned} & 12.00000 \times 0.9893 + 13.00335 \times 0.0107 \\ &= 11.8716 + 0.1391 \\ &= 12.01 \end{aligned}$$

12. (5 points)

Physical properties: melting point = 420 °C; hardness = 2.5 Mohs; density = 7.13 g/cm³ at 25 °C.

Chemical properties: granules react with dilute hydrochloric acid to produce hydrogen gas; at elevated temperatures, reacts slowly with oxygen gas to produce ZnO.

13. (3 points)

- (a) compound
- (b) heterogeneous mixture
- (c) element